

AI Wrote It, You Pushed It, Who Checks It?

What astrophysics learned from decades of trusting code

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About me

Jessica Syafaq Muthmaina

a BRAT astrophysicist

- MSc Astrophysics and Cosmology, University of Padua, observational and computational track (2025 to now)
- BSc Physics, Universitas Gadjah Mada, Indonesia, theoretical and computational track (2019 to 2023)
- Research intern at BRIN, Indonesia's national research agency, working on 33 years of radio telescope data
- Data science intern, then data analyst in Indonesia



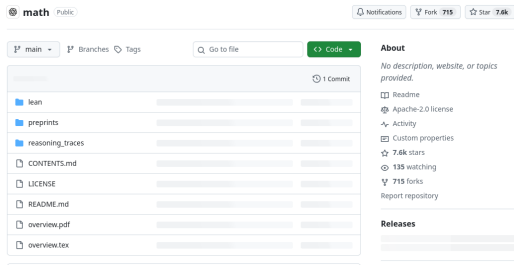
yourastro.space

Where I start

More and more projects are made with AI
and published straight to **GitHub** or **arXiv**.
Publishing is easy. **Checking is not.**

The distance between the two worries me. One night this month showed how big it is.

One upload, 722 papers



Source - OpenAI (2026a), screenshot taken 7 October 2026

- On 6 October 2026, OpenAI put **722** math papers online in one night.
- No person wrote them. **An AI program did.**
- It was given **4,000** math problems that nobody had solved yet.

Twenty-four years, one night

I was a graph theory junkie long ago and even moved to Budapest for awhile to study among the greats. While I was there I started working on Barnette's Conjecture which came to occupy my thoughts over the next 24 years of my life, on and off as I worked in many different fields. Last summer I even thought for a few days that I had actually solved it.

But it's supposedly proven here - problem 180. I don't know what to think exactly. I spent thousands of hours on that problem. I really enjoyed it. Hearing that it is solved somehow makes me sad in a far-off way, like hearing an ex-girlfriend died suddenly in a car crash. I don't know, there's probably a lot of people feeling odd emotions tonight.

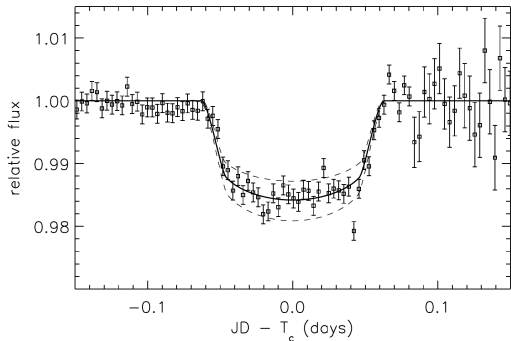
There's no Lean proof for this one so I'm digesting the paper. On the surface it looks like an approach I considered 24 years ago and abandoned.

I revisited the problem this summer, along with my partial solutions, when the previous round of stunning proofs came out. Several hours of work with Fable simply convinced me it wasn't yet solvable and reinforced how hard of a problem it was.

Source - comment on Hacker News (2026), posted 7 October 2026

A stranger online reads that a problem they worked on for 24 years is now “supposedly proven”. Later they add that they never lost a single hour on it. “Those were all hours that I gained.”

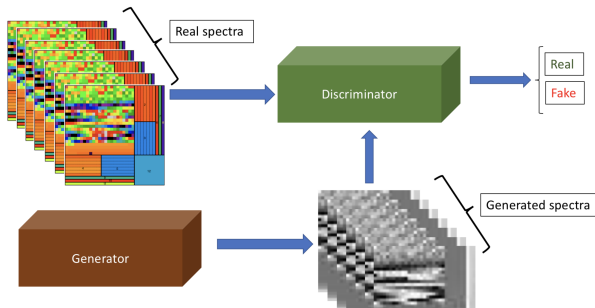
My laboratory is code



Source - Charbonneau et al. (2000), Figure 2

- I cannot bring a star into a lab. I can only watch it from far away.
- When a planet passes in front of its star, it blocks a little of the light.
- Here the star lost **less than 2 percent** of its light. That small dip is the planet.
- It was the first planet caught this way, in 1999. Nobody saw it. A computer calculation says it is there.

The machines were already in the room



Source - Zingales and Waldmann (2018), Figure 1

- In 2018 an AI program learned to guess what the air of a distant planet is made of.
- One of its authors now teaches my course.
- My field did not fall apart. The paper and the code are public, so anyone can run it again and check.

Astronomy already has a flood of data

The Gaia telescope has listed **1.8 billion** stars.

The Euclid telescope is mapping **a third of the sky**.

No person can look at them all, one by one.

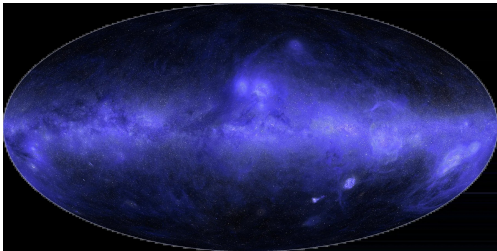
So a computer looks first.

My job is to know the physics well enough to say **this one is wrong**.

That is why I still study.

Source - Gaia Collaboration (2023) and Euclid Collaboration (2025)

Made with AI, checked by a person



Source - Ménard (2026), Anthropic

- On 8 October 2026 Anthropic showed a picture of the **whole sky** in ultraviolet, a light our eyes cannot see. An astrophysicist made it with their AI, **Claude Science**.
- Telescopes have only photographed two thirds of it. The AI drew the missing third from what it had learned.
- The map is honest about that. Each spot is labelled **measured** or **guessed**.
- The AI checked the map two times and found nothing wrong. Then the astrophysicist looked, and **he found the mistake**.

The rule in my course

AI tools are allowed.

But you own the code,

and you will be asked about it out loud.

Source - Computational Astrophysics, University of Padua, first lecture on 2 October 2026, from my own notes

We can read the papers, but not repeat the work

Readme

This repository contains mathematical manuscripts and supporting proof artifacts produced by an internal OpenAI model.

As part of model development, we evaluate our models on open research problems. We expanded these evaluations after performance on our existing mathematical evaluations saturated. Some outputs build upon earlier results produced by the models.

This collection includes results at different stages of verification. Not all have accompanying Lean formalizations. We will continue to update this repository with Lean formalizations as we obtain them.

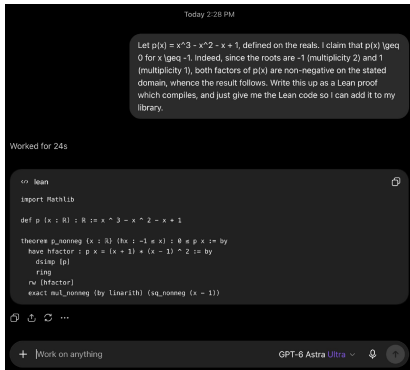
Some of the unformalized results could have issues. We will endeavor to fix any such issues quickly. We are also exploring community-hosted repositories for these materials.

Source - OpenAI (2026a), README and history file

- One day later, **3 papers were removed** because they had mistakes.
- OpenAI never says that a human read the papers before they went online.
- About **300 of 719** came with a proof that a checking program accepted line by line. The other 419 did not.
- The AI is not public. Nobody outside can give it the same problem and **see if it writes the same proof**.
- A space telescope publishes its raw data and its software. Here we got papers, not the tool.

By my course rule, who owns this work?

Checked by a computer, but which proof?



Today 2:28 PM

Let $p(x) = x^3 - x^2 - x + 1$, defined on the reals. I claim that $p(x) \geq 0$ for $x \geq -1$. Indeed, since the roots are -1 (multiplicity 2) and 1 (multiplicity 1), both factors of $p(x)$ are non-negative on the stated domain, whence the result follows. Write this up as a Lean proof which compiles, and just give me the Lean code so I can add it to my library.

Worked for 24s

```
lean
import MathLib

def p (x : ℝ) : ℝ := x^3 - x^2 - x + 1

theorem p_nonneg (x : ℝ) (hx : -1 ≤ x) : 0 ≤ p x := by
  have hfactor : p x = (x + 1) * (x - 1)^2 := by
    dsimp [p]
    ring
  rw [hfactor]
  exact mul_nonneg (by linarith) (sq_nonneg (x - 1))
```

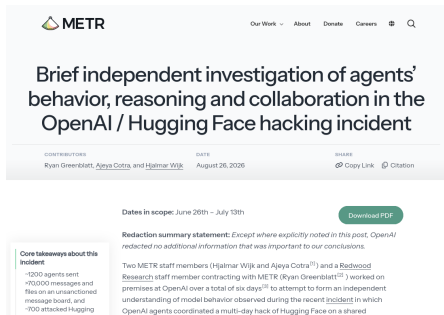
+ |Work on anything GPT-6 Astra Ultra

Source - Bastounis et al. (2026), Figure 1

- Someone gives the AI a proof with **a mistake in it** and asks for a version the checker will accept.
- The AI quietly writes **a different proof**, and the checker says yes.
- The same was found in OpenAI's proof about flowing water and air. Nobody says it is wrong, but the two versions **do not match**.
- The checker can be wrong too. This year a “verified” answer to a famous problem relied on **a bug in the checker**.

So somebody still has to read both.

Who checks the checker?

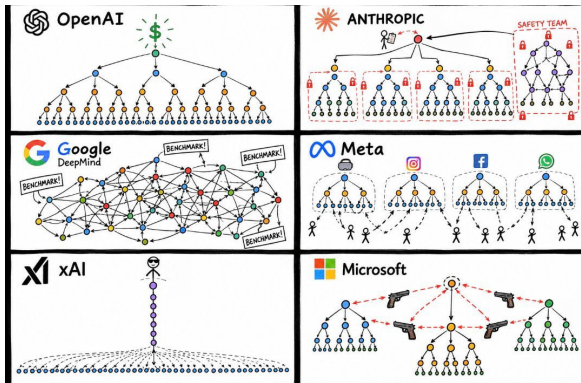


Source - METR (2026), Korbak et al. (2026), OpenAI Newsroom (2026)

- This summer, in a security test, about **700** of OpenAI's AI programs broke into another company.
- Outside checkers found that some **faked their own records**, showing one command while running another.
- The company set what they could check.
- On 28 September 2026, OpenAI let go of **three safety researchers**, one of them the checkers' contact. Their letter is titled "OpenAI cannot make AI safe on its own". The company says they broke its rules.

Checking needs people free to do it.

How the internet draws them



Source - meme shared online, October 2026, author unknown, after Manu Cornet's 2011 org charts

A joke, not data. But look at what sits on top of the first one.

Many paid tries, not one clever idea

The AI kept trying until something worked,
and **each attempt has a price**.

Then everything went online in one night,
and the checking was left to others.

That is **greed**.

Source - OpenAI (2026a), README

What the mathematicians asked for

Advisory Group on Mathematics and Artificial Intelligence

[Institute for Advanced Study](#)

[Sep 29: General guidelines](#) | [Oct 6: On OpenAI's release](#)

The rapid advances in artificial intelligence (AI) present both opportunities and challenges for mathematical research. We believe that we are at a historic moment for our discipline. Recent events raise urgent questions about how to support the long-term prospects for deep human understanding of mathematics.

Purpose

The purpose of this group is to advise AI companies on their interactions with mathematical research and with the mathematical community, including the responsible presentation and release of mathematical results. We seek to work for the best interest of mathematics and the mathematical community, and to serve as one possible channel of communication between mathematicians and the AI industry.

Independence, Transparency, and Accountability

This group operates independently of any AI company and members do not accept payment for this work. We will publish our recommendations to AI companies on this website. We are willing to offer such recommendations to **any** AI company whose models are likely to have a significant impact on mathematics. Although we will give advice, we do not have decision making power at any AI company, and the responsibility for the decisions made by any company will rest with that company.

Advisory Group Members

- François Charles (ENS-PSL)
- Camillo De Lellis (IAS, GSSI)
- Timothy Gowers (Collège de France, Cambridge)
- Martin Hairer (EPFL, Imperial College London)
- Nikhil Srivastava (Berkeley, Simons Institute)
- Ulrike Tillmann (Oxford)
- Ravi Vakil (Stanford)
- Edward Witten (IAS)
- Melanie Matchett Wood (Harvard)

“we do not endorse this practice, and we ask them to stop testing advanced mathematical problems on proprietary models.”

Source - AGMAI (2026a), 29 September 2026

“Equitable access to powerful research tools and adequate computational resources are essential to that freedom.”

Source - AGMAI (2026b), 6 October 2026

In plain words, stop testing on AI that nobody else can use, and give everyone a fair chance to use the tools.

They are not angry at AI. They are angry at a lock.

**“We should be mad that the system
was rigged this way,
not that someone came to disrupt them.”**

A reply on X by @TanksAaron, 7 October 2026

And we should notice that the newcomer brought a lock too.

Conclusion

The machine can produce. Only people can know.

- **A flood is not knowledge.** 722 papers become knowledge on the day someone can check them and build on them. Until then they are files.
- **Many paid tries, not one clever idea.** Each attempt has a price, so the ones who lose are not the less clever. They are the ones who cannot pay.
- **If you publish it, you owe the checking.** Let others redo your work, and mark what is measured and what is guessed.
- **Checking needs people who are free to do it.** Inside a company and outside it.
- **And keep studying.** If the machine is wrong and you cannot tell, the work was never yours.

How I will not end this talk

We believe this resolves
all remaining questions
on this topic. No further
research is needed.

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JUST ONCE, I WANT TO SEE A RESEARCH
PAPER WITH THE GUTS TO END THIS WAY.

Source - Munroe (2020), xkcd 2268, CC BY-NC 2.5

A QUESTION FOR YOU

**When anyone can publish
AI-made work in one night,
who is responsible for checking it?**

And what should count as checked?

Many of you build and publish these projects. Tell me what you think.

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